

Aave V3 Liquidator — Command Cheat Sheet

Base · Event-purity stack (Flashblocks WS + oracle bootstrap + shadow validator) · VM quick reference

Current VM startup sequence (recommended)

One-time setup

Step	Command
Enter repo	<code>cd ~/optimism-aave-v3-liquidator-ts</code>
Install deps	<code>npm install</code>
Create env profiles	<code>npm run env:bootstrap</code>
Sync template keys	<code>npm run env:merge-example -- --target .env.event-purity-soak</code> <code>npm run env:merge-example -- --target .env.event-purity-production</code>
Make executable	<code>chmod +x start-*.sh watch-bot.sh scripts/launcher-run-bot-detached.sh</code>
Edit secrets	Edit <code>.env.event-purity-soak</code> and <code>.env.event-purity-production</code>

Before first live run

Step	Command	Notes
A	<code>./start-event-purity-soak.sh</code>	48h shadow soak, no live TX
B	<code>./watch-bot.sh</code>	Monitor; <code>--once</code> for snapshot
C	<code>npm run dry-run:receipt:event-purity-production</code>	Required for safety gate
D	<code>./start-event-purity-production.sh</code>	Start live within 15 min of receipt

What each start script does

- `npm run bot:stop`
- `node scripts/ensure-redis.mjs`
- `npm run build`
- `node scripts/preflight-event-purity-env.mjs` (event-purity only)
- `scripts/launcher-run-bot-detached.sh` — detached `nohup bot`

Logs: `logs/<prefix>-YYYYMMDD-HHMMSS.log` · Pointer: `logs/latest-session.txt`

Day-to-day ops

Action	Command
Monitor (30s refresh)	<code>./watch-bot.sh</code>
Single snapshot	<code>./watch-bot.sh --once</code>
Monitor + audit	<code>./watch-bot.sh --audit</code>
Tail log	<code>tail -f logs/event-purity-production-*.log</code>
Process status	<code>npm run bot:status</code>
Stop	<code>npm run bot:stop</code>
Health	<code>curl -s http://127.0.0.1:9090/healthz</code>
Status JSON	<code>curl -s http://127.0.0.1:9090/status</code>
Metrics	<code>curl -s http://127.0.0.1:9090/metrics</code>

VM shell launchers (Ubuntu/Linux)

Command	Env profile	Mode
<code>./start-event-purity-soak.sh</code>	<code>.env.event-purity-soak</code>	Recommended soak
<code>./start-event-purity-production.sh</code>	<code>.env.event-purity-production</code>	Recommended live
<code>./start-simulation.sh</code>	<code>.env.simulation</code>	Legacy soak
<code>./start-production.sh</code>	<code>.env.production</code>	Legacy live
<code>./start-production-budget.sh</code>	<code>.env.production.budget</code>	Legacy live (RPC budget)
<code>./watch-bot.sh [--once --audit --interval N]</code>	—	Live monitor

Windows launchers (.cmd)

File	Purpose
Start Event-Purity Soak.cmd	48h event-purity soak
Start Event-Purity Production.cmd	Event-purity live
Start Simulation Bot.cmd	Legacy simulation
Start Production Bot.cmd	Legacy live
Start Production Bot (Budget).cmd	Legacy live, RPC budget
Start Production Bot (No Gate).cmd	Legacy live, skip safety gate
Start Live Bot.cmd	Foreground ts-node live
Stop Bot.cmd	Stop bot
Setup Dry Run Receipt.cmd	Dry-run receipt helper

npm — bot lifecycle

Command	Purpose
<code>npm run build</code>	Compile TypeScript
<code>npm run start</code>	Run compiled bot
<code>npm run dev</code>	ts-node dev run
<code>npm run start:sim</code>	Foreground simulation
<code>npm run start:live</code>	Foreground live
<code>npm run bot:status</code>	Lock + PID status (JSON)
<code>npm run bot:stop</code>	Graceful stop
<code>npm run redis:ensure</code>	Start/verify Redis

npm — environment

Command	Purpose
<code>npm run env:bootstrap</code>	Create all env profiles
<code>npm run env:merge-example</code>	Merge <code>.env.example</code> keys
<code>npm run dry-run:receipt</code>	Receipt for legacy production
<code>npm run dry-run:receipt:event-purity-production</code>	Receipt for event-purity live

npm — test & validation

Command	Purpose
npm test	Unit tests (vitest)
npm run test:watch	Vitest watch
npm run test:coverage	Coverage
npm run test:chaos	Chaos tests only
npm run typecheck	TypeScript check
npm run test:flashblocks	Flashblocks WS probe
npm run verify:pre-soak-phase0	Pre-soak phase-0 check
npm run benchmark:base	Base replay benchmark
npm run benchmark:rpc	RPC latency benchmark
npm run check:rpc-gates	RPC SLO gates
npm run probe:multicall-base	Multicall batch probe
npm run debug:hf-sweep-base	HF sweep debug
npm run replay:historical	Historical replay
npm run replay:liquidation-block	Liquidation block replay

npm — audits & reports

Command	Purpose
npm run audit:watchlist-base	Watchlist audit
npm run audit:dex-surface-base	DEX surface audit
npm run audit:morpho-preliquidation-base	Morpho audit
npm run report:competitive-gap	Competitive gap (48h)
npm run review:daily-pipeline	Daily pipeline review
npm run pairs:top-base	Top Base pairs

npm — build & deploy

Command	Purpose
npm run compile:contracts	Compile Solidity receiver
npm run deploy:liquidation-receiver:base	Deploy on Base
npm run rust:build	Build rust-hotpath

Direct node scripts

Command	Purpose
node scripts/audit-session.mjs <log>	Full session audit
node scripts/watch-bot-summary.mjs <log>	Session summary
node scripts/detect-critical-log-errors.mjs <log>	Critical error scan
node scripts/preflight-event-purity-env.mjs <env>	Env preflight
node scripts/verify-bot-launch.mjs <log>	Verify launch in log
node scripts/soak-production-report.mjs	Soak report
node scripts/secret-scan.mjs	Secret scan

PM2 (optional alternative)

Command	Purpose
<code>pm2 start ecosystem.config.cjs</code>	Start via PM2
<code>pm2 logs aave-liquidator-base</code>	Tail logs
<code>pm2 stop aave-liquidator-base</code>	Stop

Troubleshooting

Symptom	Action
Live blocked	Run dry-run receipt; start within 15 min
Missing profile	<code>npm run env:bootstrap</code>
Multiple bots	<code>npm run bot:status</code> then <code>npm run bot:stop</code>
Sequencer bootstrap fatal	Wait for Base sequencer recovery + grace

See [docs/ENV_PROFILES.md](#) and [README.md](#) for full detail.